

Anant Goyal

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EDUCATION

University of Illinois Urbana-Champaign, *Masters in Computer Engineering*

GPA: 3.6/4.0 | Dec 2026

The University of British Columbia, *Bachelors in Computer Engineering*

GPA: 3.5/4.0 | May 2023

EXPERIENCE

Software Engineer, *Citi*

Jul 2023 - Aug 2025

- Reduced trade creation time by 90% for high-volume institutional users by owning and shipping a bulk trade feature for a high-throughput money market fund trading application in Java and Angular
- Designed financial safeguards such as duplicate trade detection, blind re-key for trades over \$5M, maximum trade amount of \$10B, and multiple approval flow for trades over \$10M to ensure compliance and risk reduction
- Cut QA cycle time by 25% by spearheading unit test infrastructure, driving code coverage from 0% to 80%
- Developed a containerized test-automation microservice on Amazon ECS using Selenium Grid, distributing 400+ regression tests and cutting execution time by 62%
- Mentored incoming engineers and 2 interns through onboarding by scoping starter tickets, conducting code reviews, and leading architectural discussions with operations stakeholders

Software Engineer Intern, *Picotera Electronics*

May 2022 - Aug 2022

- Increased device uptime by 10% and improved emergency alerts functionality (pedestrian-vehicle safety, duplex communication) by developing firmware in C for a smart communication device
- Delivered custom Android app features such as app-whitelisting and kiosk mode using the Android Management API to manage device policies for over 50 devices used by customers

Software Test Engineering Intern, *NETGEAR*

Jan 2021 - Aug 2021

- Developed and executed automated test scripts in Python for 3G/LTE routers across multiple product lines
- Diagnosed low-level network issues across TCP/IP, analyzing packet traces during router testing using Wireshark

Test Automation Development Intern, *IBM*

May 2020 - Dec 2020

- Automated 30+ regression test cases for IBM Cognos Analytics, a large-scale enterprise BI platform, reducing manual testing load by 40% using Selenium in JavaScript
- Improved i18n validation coverage by 100% across localized environments by developing a CSV parser tool to enable test execution across 20+ supported languages

PROJECTS

Rex App | *React Native, TypeScript, Expo, Docker, Postgres, GCP, PostHog, Sentry, Claude Code*

[Demo](#)

- Built Rex, a social discovery app currently in beta with early users, to rank places through pairwise comparisons, using a binary-search algorithm bounding user effort to $O(\log n)$ comparisons per new entry
- Cut data egress by 7x, implementing client-side image compression, public CDN caching and feed pagination

Distributed File System & Stream Processor | *Java, Hadoop, Apache Spark*

[Github](#)

- Designed and built a fault-tolerant, stateful distributed file system and stream processor achieving 10,000 tuples/sec throughput and 70ms re-replication latency under node failures across a 10-node cluster
- Implemented leader-based coordination with consistent hashing and quorum-based reads/writes to ensure strong consistency and automatic rebalancing under varying workloads improving performance

Load Generator Observability Tool | *Go, Python, JMeter, k6, Prometheus, Grafana*

[Paper](#)

- Discovered open-loop load generators silently under-report p90 tail latency by up to 32% under realistic load, caused by client-side CPU and connection bottlenecks invisible to aggregate metrics
- Built an observability layer to diagnose load generator failures into root-cause categories; achieves 98% classification accuracy across fault-injection experiments on DeathStarBench microservices; paper submitted to ACM SoCC 2026

SKILLS

Programming: Java, Python, C/C++, C#, Go, TypeScript, JavaScript, SQL, Bash

Technologies: React, Angular, Spring, Hadoop, Spark, Next.js, Node.js, Postgres, Playwright, Vitest, Selenium

Tools: AWS, GCP, Git, GitHub Actions, Jenkins, Snyk, PostHog, Autosys, Jira, Claude Code, Gemini API